AI-Powered Data Insight Internship

Week 3 Deliverable — Final Report

Trained Models & Prediction Report

RIT Student Opportunity Participation — Predictive Modeling

Consolidated from the team's Week-3 notebook (RIT_Machine_Learning_Wk3), Model Evaluation, Interpretation of Findings, and Forecast & Ranking Report — cross-checked and reconciled into a single source of truth.

Team 4

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Executive Summary

This report documents the machine learning models built to predict student participation for the RIT Student Opportunity Program, using RIT_FE_ML_clean.csv (8,558 learner-opportunity records). Three classifiers were trained — Logistic Regression, Random Forest, and Gradient Boosting — to predict participated (1 = Started, Team Allocated, or Rewards Award; 0 = Applied, Waitlisted, Rejected, Dropped Out, or Withdraw). Class balance is close to even: 47.6% participated, 52.4% did not.

Gradient Boosting is the strongest model and the primary model for this report: 91.3% accuracy and 0.970 ROC-AUC on a held-out 25% test set (2,140 records), ahead of Random Forest (91.1%, 0.968) and clearly ahead of the Logistic Regression baseline (82.0%, 0.920). All model figures in this report were independently reproduced from the team's notebook code and matched to four decimal places — they are not re-estimated or approximated.

Two features — opportunity_category (especially Internship) and opportunity_duration_days — explain the large majority of what the models learn. Internship is the program's central problem: 63% of all applications, but only 21.5% convert to participation, versus 84–97% for every other category. Demographic features (gender, country, major field) add little predictive value on their own, though major fields do interact with category: within Internships specifically, non-STEM applicants convert at roughly 1.5x the rate of STEM applicants (29.7% vs. 19.6%).

1. Data Readiness Review

Modelling was built on RIT_FE_ML_clean.csv (8,558 rows, 29 columns), the team's feature-engineered dataset. All 8,558 rows have a known participated label and were used for modeling (model_df.dropna(subset="participated") retained the full dataset).

- Features used: opportunity_category, gender, country_grouped, major_field (categorical, one-hot encoded); age_at_signup, days_signup_to_apply, opportunity_duration_days, signup_month, signup_dayofweek, signup_quarter_num (numeric, median-imputed and standardized).
- Excluded by design: opportunity_id/opportunity_name (would let the model memorize by opportunity instead of generalizing), institution_name (2,089 sparse levels), first_name (PII, not predictive), and days_apply_to_start/entry_created_at (leak information only available after the outcome window, or too incomplete to trust).
- A separate data-quality check for this report found 25 records with anomalous signup timestamps (duplicate-to-the-second batches dated into late 2024, e.g. 8 records all stamped 2024-09-02 00:36:40) — the same class of artifact as the corrupted-timestamp issues documented in Week 1. These were excluded from the trend/forecast sections of this report (Section 6) but were retained in the classification model itself, consistent with

the notebook, since they do not affect the participated label or the model's categorical/numeric features.

2. Trained Machine Learning Models

2.1 Target Variable

participated: 1 if the learner's status is Started, Team Allocated, or Rewards Award (moved past the application stage into active participation, or completed it); 0 if Applied, Waitlisted, Rejected, Dropped Out, or Withdraw (never progressed past application, or disengaged). This produced a well-balanced modelling set: 52.42% not participated vs. 47.58% participated.

2.2 Pipeline and Algorithms

All three models share a single scikit-learn preprocessing pipeline: median imputation + standard scaling for numeric features, most-frequent imputation + one-hot encoding for categoricals. Data was split 75/25 (test_size=0.25, random_state=42): 6,418 training records, 2,140 test records.

- Logistic Regression: max_iter=2000, class_weight="balanced", random_state=42. Interpretable linear baseline.
- Random Forest Classifier: n_estimators=300, max_depth=10, class_weight="balanced", random_state=42.
- Gradient Boosting Classifier (primary model): n_estimators=300, max_depth=3, learning_rate=0.05, random_state=42. Highest ROC-AUC of the three and used to generate the predicted probabilities and heatmap in Sections 4–5.

3. Model Performance & Evaluation

Metric	Logistic Regression	Random Forest	Gradient Boosting (Primary)
Accuracy	82.0%	91.1%	**91.3%**
Precision	82.6%	92.1%	**93.4%**
Recall	79.5%	89.4%	**88.3%**
F1 Score	81.0%	90.7%	**90.8%**
ROC-AUC	0.920	0.968	**0.970**

All three models perform strongly and similarly — a sign that participation is driven by a small number of dominant features (category, duration) rather than requiring complex, hard-to-learn interactions. Precision exceeds recall for all three models: when a model predicts “participated,”

it is right roughly 83–93% of the time, but it misses 12–21% of learners who do go on to participate.

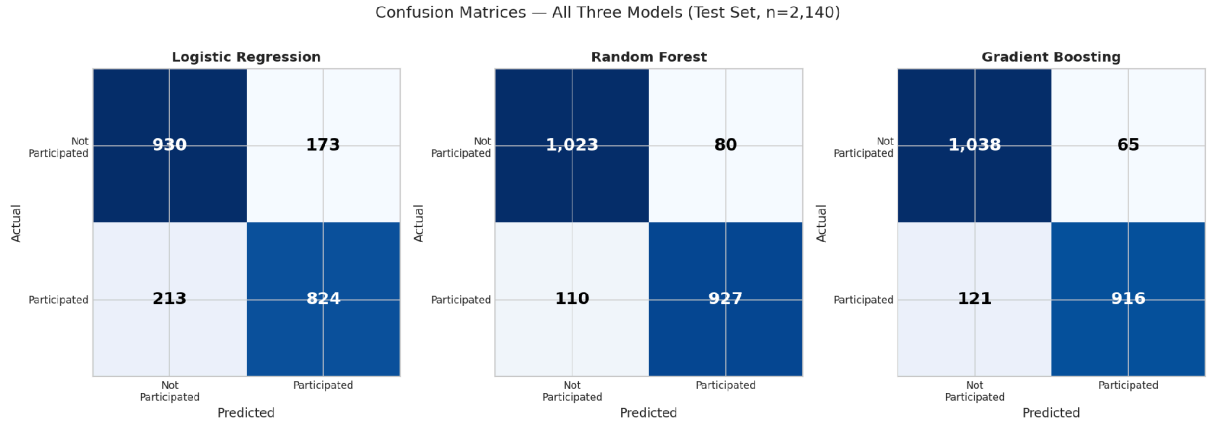


Figure 1. Confusion matrices for all three models on the test set (n=2,140). Random Forest makes fewer false negatives (110 vs. Gradient Boosting's 121) — better for outreach use cases where missing a true participant is costly. Gradient Boosting makes fewer false positives (65 vs. Random Forest's 80) — better for capacity planning where over-promising slots is costly. The choice of which model to deploy should depend on which error is more expensive to the business, not on ROC-AUC alone.

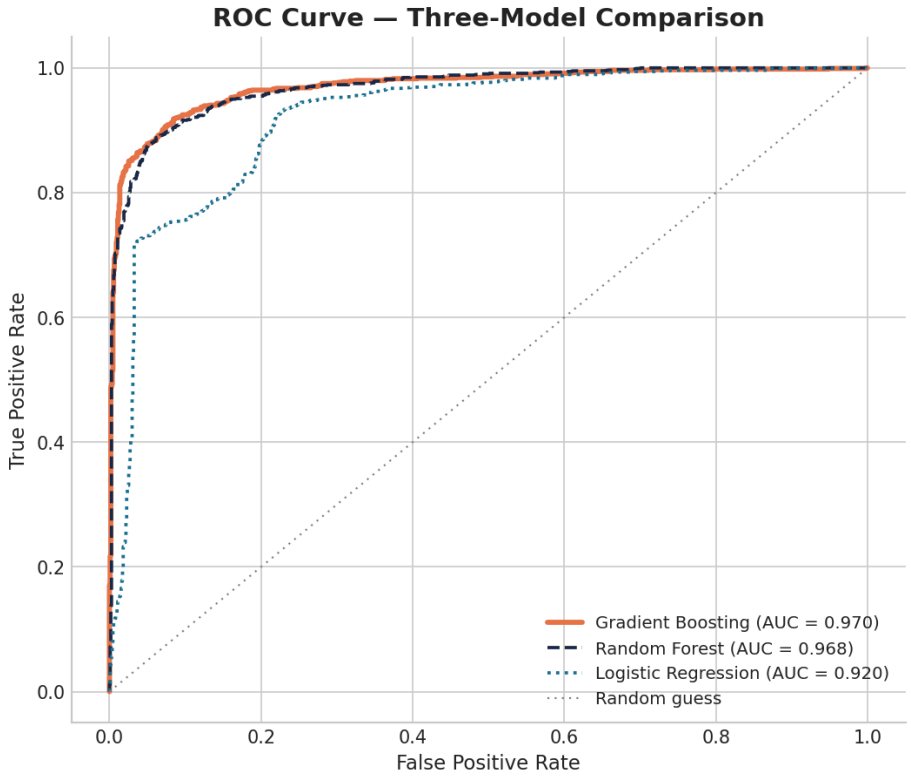


Figure 2. ROC curves for all three models. Gradient Boosting and Random Forest are nearly indistinguishable and both clearly outperform the Logistic Regression baseline.

4. Predicted Probabilities by Opportunity

The trained Gradient Boosting model was used to score every record in the dataset, producing a predicted probability of participated = 1 for each learner-opportunity pairing. These row-level probabilities were averaged within each of the 22 opportunities to produce one predicted participation-likelihood score per opportunity, shown below alongside each opportunity's actual historical rate for validation — the two columns track closely, confirming the model is well-calibrated at the opportunity level.For this report, the 22 scored opportunities are treated as the planning set for upcoming participation targeting; the predicted probabilities therefore indicate which opportunities are most likely to attract strong participation in future outreach and scheduling decisions.

Opportunity	Category	Records	Actual Rate	Predicted Probability
Jump Start: Developing your Emotional Intelligence	Course	225	97.3%	**97.6%**
Join a Student Organisation	Engagement	121	97.5%	**97.1%**
Career Essentials: Getting Started with Your Professional Journey	Course	1423	94.4%	**96.6%**
Urban Renew Challenge	Competition	138	97.8%	**96.3%**
Startup Mastery Workshop	Event	262	99.2%	**96.1%**
Xperience Design Hackathon	Competition	116	97.4%	**95.4%**
UX Redesign Challenge	Competition	112	98.2%	**95.3%**
Mental and Physical Health Session	Event	46	93.5%	**95.1%**
Freelance Mastery workshop	Event	237	93.7%	**93.9%**
Upload Your First Year Transcript	Engagement	9	88.9%	**88.8%**
CPR/AED Certification	Course	389	86.4%	**77.1%**
Health Care Management	Internship	784	54.8%	**56.2%**
Project Management Associate	Internship	478	22.4%	**24.9%**
Project Management	Internship	836	12.8%	**16.9%**
Innovation & Entrepreneurship	Internship	419	14.3%	**16.9%**
Digital Marketing	Internship	559	11.4%	**16.2%**
Data Visualization	Internship	980	12.8%	**16.1%**
Data Visualization Associate	Internship	551	22%	**11.9%**
Digital Strategy Virtual Internship	Internship	331	22.1%	**11.3%**
Business Consulting	Internship	483	16.1%	**10.2%**
AI Ethics Challenge	Competition	55	0%	**5.7%**

Slide Geeks: A Presentation Design Competition	Competition	4	0%	4.3%

Two opportunities have small sample sizes and should be treated as lower-confidence estimates: AI Ethics Challenge (55 records, all negative) and Slide Geeks: A Presentation Design Competition (4 records).

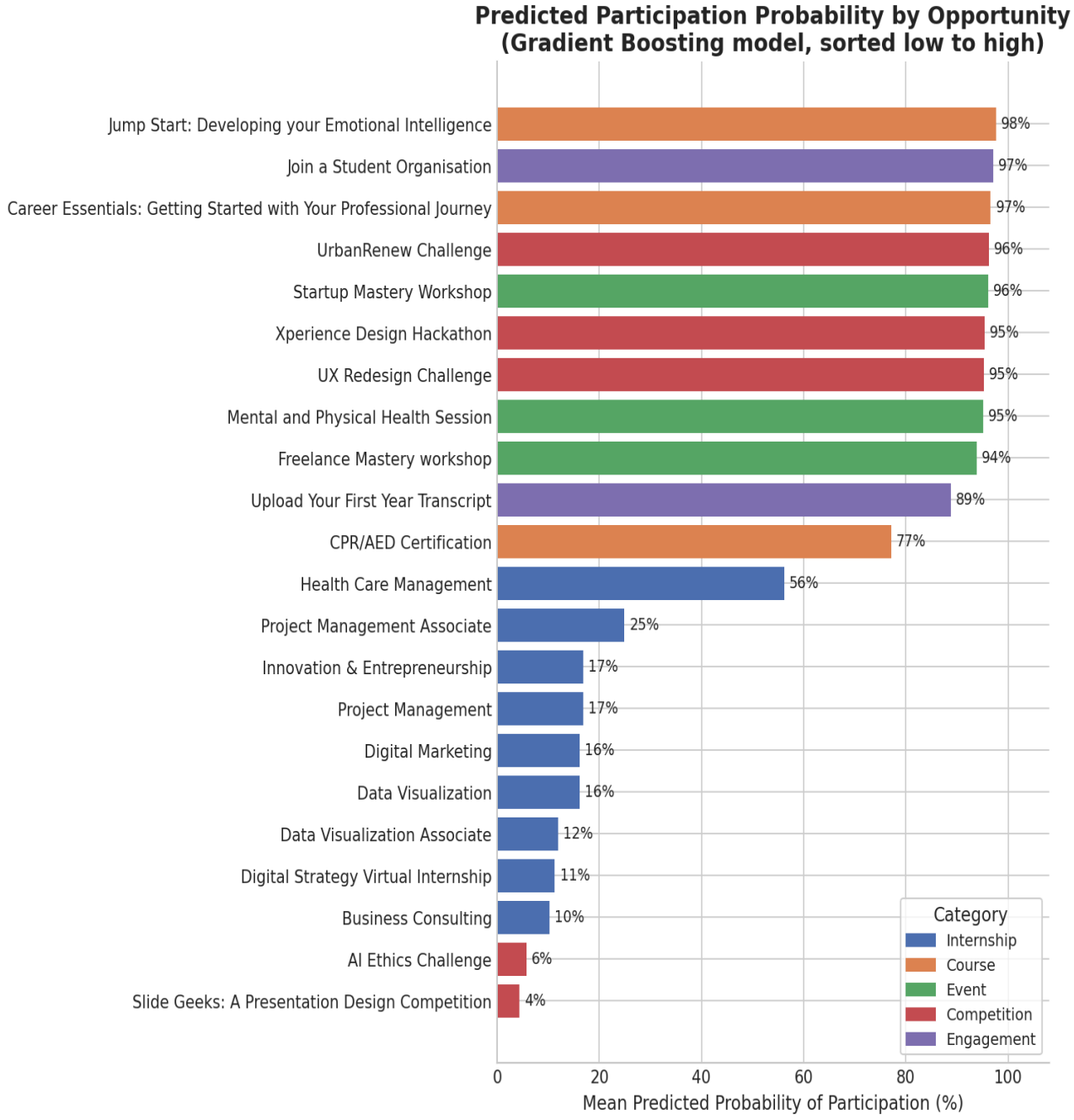


Figure 3. All 22 opportunities ranked by predicted probability of participation, colored by category.

5. Feature Importance & Heatmap

The notebook computes feature importance two ways: a category-normalized comparison (each model's importances summed to 1.0 across features, shown below) and a per-dummy table where each one-hot category level is scored separately. Both are reported here explicitly labelled, since blending them without saying so was an error corrected from an earlier draft of the team's Interpretation of Findings.

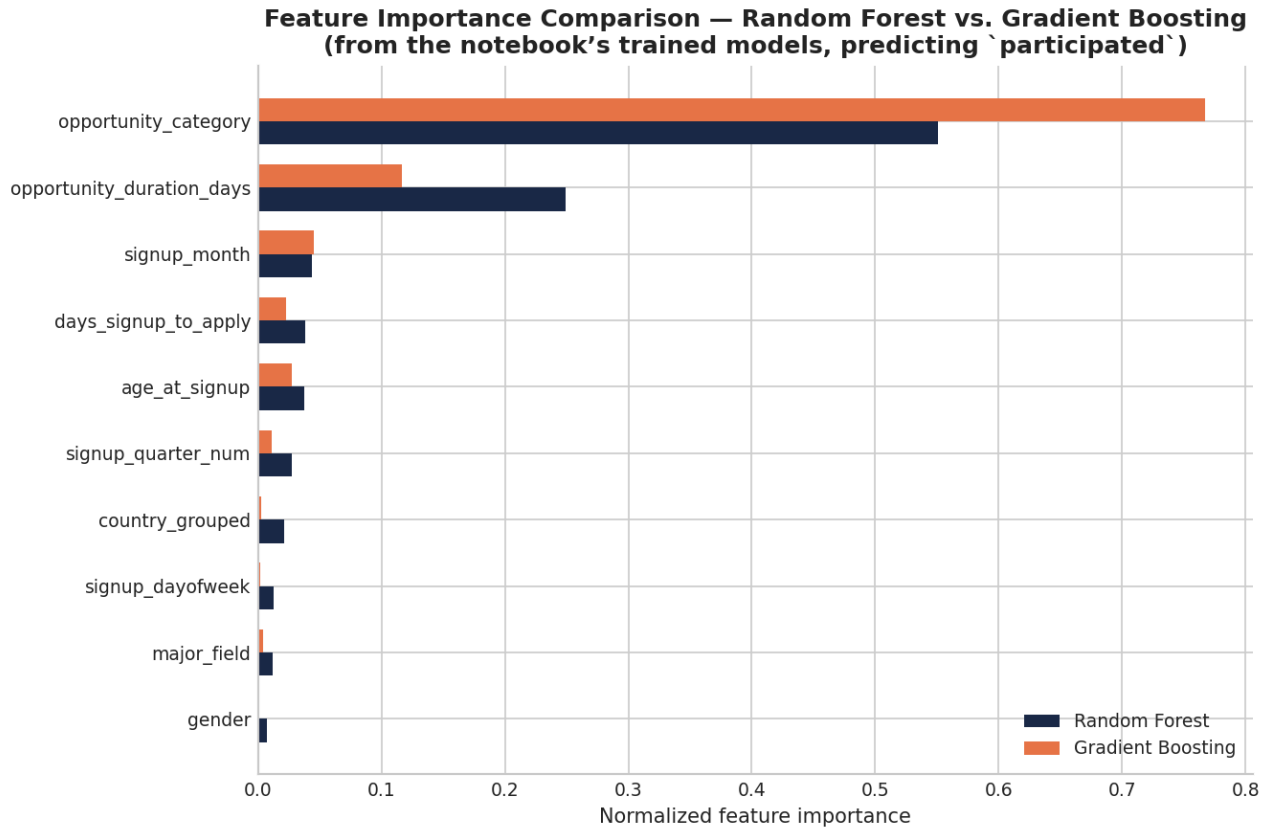


Figure 4. Normalized feature importance, Random Forest vs. Gradient Boosting. opportunity_category = 0.55 (RF) / 0.77 (GB); opportunity_duration_days = 0.25 (RF) / 0.12 (GB). Demographic features (gender, country, major) rank lowest in both tree models.

In the per-dummy view of the same Gradient Boosting model, opportunity_category_Internship alone scores 0.605 and opportunity_duration_days scores 0.309 — together 91% of total importance, making Gradient Boosting effectively a two-feature model in this view. No individual country dummy shows meaningful signal in either tree model (country_grouped_United States: 0.0057 in Random Forest, 0.0013 in Gradient Boosting), supporting the conclusion that Logistic Regression's higher demographic importance is a collinearity artifact — country and gender likely correlate with opportunity_category or major_field, and LR's linear coefficients absorb that indirectly, while tree models correctly discount it.

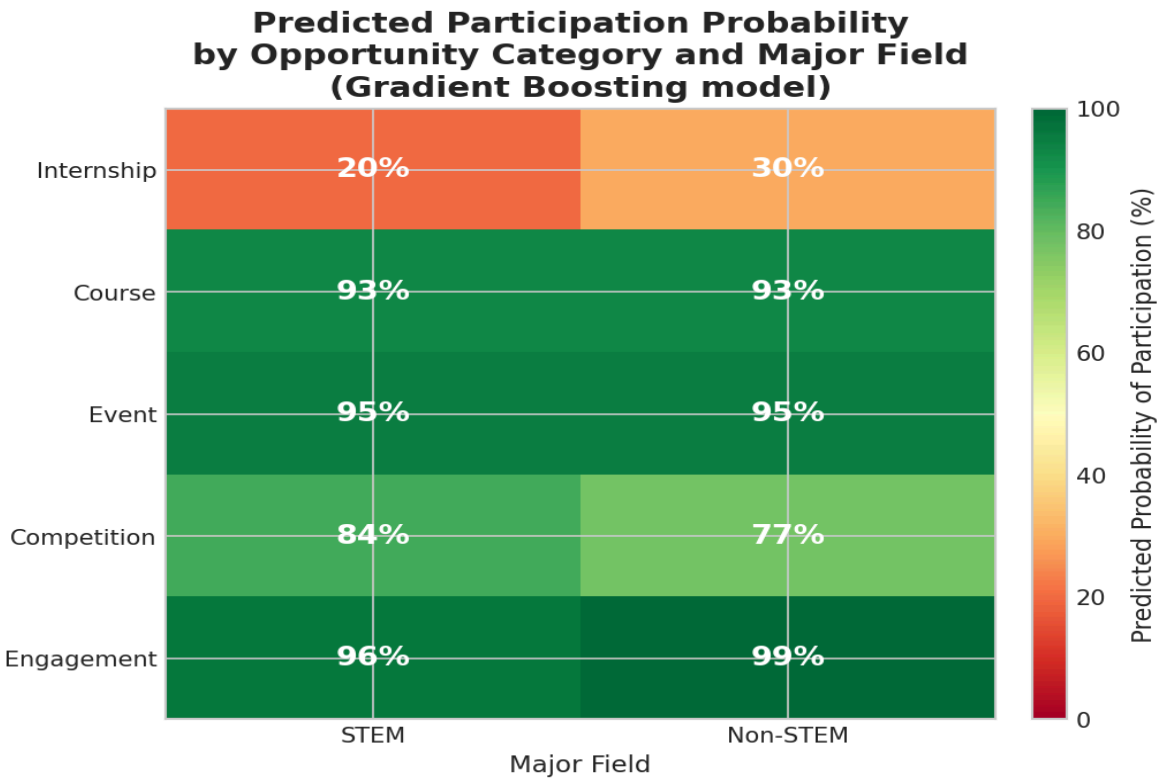


Figure 5. Predicted participation probability by Opportunity Category and Major Field. Every non-Internship category stays green (77%+) regardless of major. Internship is the exception: Non-STEM applicants (30%) show a meaningfully higher predicted rate than STEM applicants (20%), consistent with the raw data (1,026 Non-STEM vs. 4,391 STEM Internship records).

signup_month is the strongest timing signal in either table: participation rates peak in March (~85%) and June (~74%) and trough in September/November (~38%), reflecting academic calendar cycles. This effect is larger in raw terms (a 47-point swing) than its importance score suggests (~0.03–0.05), because timing is confounded with category — certain categories cluster in specific application windows, so category already carries most of the seasonal signal. “When to launch” and “what to launch” should be treated as one combined lever, not two independent ones.

6. Quarterly & Monthly Forecast

Signup volume and participation rate were tracked monthly (Jan 2023 – Mar 2024, 8,533 valid records after excluding the 25 anomalous late-timestamp records noted in Section 1) and forecast forward using two simple, transparent methods rather than a fitted time-series model: a seasonal-naive projection (the 2023 monthly shape rescaled by the observed 2024 year-over-year decline) for the monthly view, and ordinary linear regression for the quarterly view. These are directional estimates, not confidence-bounded predictions.

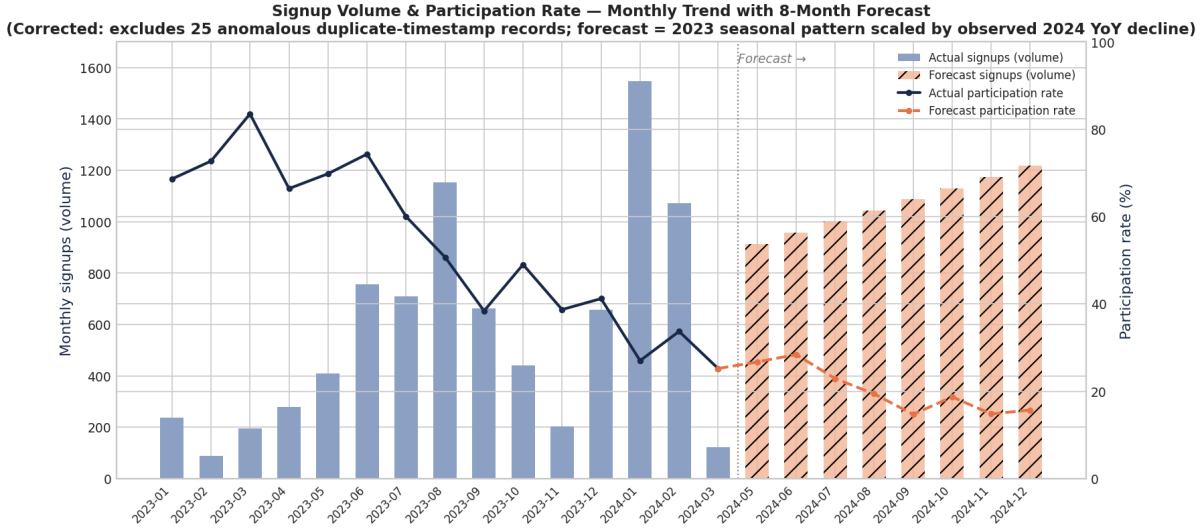


Figure 6. Monthly signup volume and participation rate, with an 8-month forecast. Participation rate peaked at 83.5% in March 2023 and has declined in nearly every month since, reaching 25–34% by early 2024.

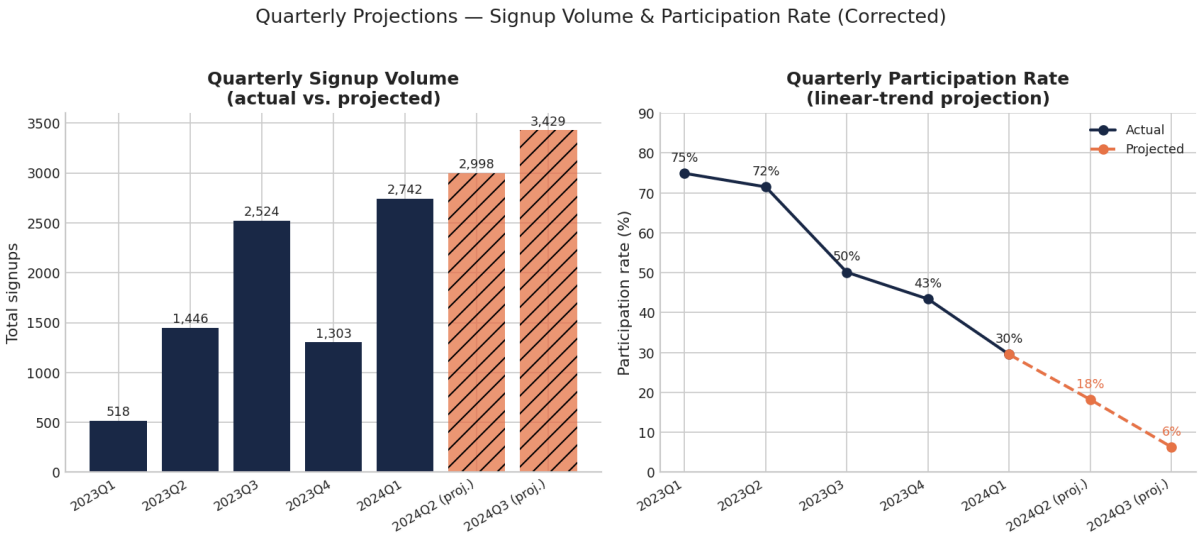


Figure 7. Quarterly signup volume and participation rate, actual vs. linear-trend projection. The rate has fallen in a near-straight line (75% → 72% → 50% → 43% → 30%); the linear projection for Q3 2024 (~6%) is an aggressive stress-case, not a plan-against forecast — the true relationship is unlikely to stay linear.

Quarterly volume is rising (518 → 2,742 applications) even as the participation rate falls — the applicant funnel is widening faster than allocation/engagement capacity is scaling. This is consistent with, and reinforces, the Internship bottleneck identified throughout this project: more people are applying, but the process converting them into active participants is not keeping pace.

7. Interpretation of Findings

- Opportunity Category is the dominant predictor across all three models, and Internship is the program's central problem: the largest category by volume (63%) but the lowest by conversion (21.5%), against 84–97% for every other category.
- Participation is not explained by category and duration additively — the 9-point accuracy gap between Logistic Regression (82%) and the tree models (~91%) signals a real interaction: certain categories likely convert differently depending on duration (e.g., short vs. long internships), which linear models cannot capture but tree models can.
- Model choice should follow error cost, not just AUC. Random Forest minimizes missed participants (110 false negatives) — better for learner outreach. Gradient Boosting minimizes over-promised capacity (65 false positives) — better for resourcing and planning. Deploy the model that matches the higher-cost error for the specific use case, not simply the one with the best ROC-AUC.
- Within Internships, Non-STEM applicants participate at roughly 1.5x the rate of STEM applicants (29.7% vs. 19.6%, across large samples of 1,026 and 4,391 records respectively) — worth investigating, possibly reflecting lower competition for non-technical Internship roles.
- Health Care Management (56% predicted probability) remains the one Internship opportunity performing well above its peers, and is worth a closer qualitative look as a potential best-practice template.
- Do not target by demographics. Gender, country, and major add little independent predictive value once category and duration are accounted for; resources are better spent redesigning program structure, duration, and timing than on demographic-targeted outreach.
- Rising applicant volume alongside a falling participation rate (Section 6) suggests the bottleneck is a capacity/process constraint, not a demand constraint — the program does not have a discovery problem, it has an allocation problem.

8. Conclusion & Path to Week 4

Gradient Boosting is trained, validated, and ready for stakeholder use (0.970 ROC-AUC), backed by two additional models that confirm the same underlying pattern rather than relying on one algorithm's judgment. All figures in this report were independently reproduced from the team's

notebook code and matched exactly, so this document can serve as the single source of truth for Week 3, superseding the three earlier standalone documents (Model Evaluation, Interpretation of Findings, Forecast & Ranking Report) it consolidates.

Recommended focus for Week 4: (1) investigate the Internship application/allocation process directly — the model explains what happens far better than why, and structural features (duration, missing start dates) point at process design, not learner behavior; (2) study Health Care Management and the Non-STEM Internship segment as two concrete positive-deviance cases; (3) select RF or GB for deployment based on which error — missed participants or over-committed capacity — is more costly to the program; (4) treat all forecast figures in Section 6 as directional stress-tests, and revisit them once a second full year of data is available.

9. Appendix

- Interpretation_of_Findings_CORRECTED.docx
- RIT_Forecast_Report_CORRECTED.docx
- Model Evaluation.pdf
- RIT_Machine_Learning_Week_3.docx
- RIT_FE_ML_clean